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Bio-Inspired Navigational Intelligence A Cross-Species Computational Framework from Mammals to Insects

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***Abstract*—Autonomous navigation in perceptually-degraded environments remains a critical challenge for artificial agents, particularly in dynamic real-world scenarios. Current deep reinforcement learning (DRL) methods often struggle with generalization, requiring extensive data and failing to maintain performance under sparse rewards and high sensory noise. This work addresses this gap by postulating that the evolutionary convergence in biological navigation systems reveals computationally efficient and robust algorithms suitable for AI implementation. We present an integrative computational framework that translates the core navigational principles of two evolutionarily distant species dogs (Canis familiaris) and carpenter ants (Camponotus pennsylvanicus) into novel, high-performance network architectures.The core contribution of this work lies in formalizing a shared core algorithm for path integration based on Bayesian inference and vector-based goal encoding. We introduce two specific neurally-plausible architectures: (1) a Sparse Olfactory-Grid LSTM, which integrates sparse olfactory-spatial processing inspired by the canine dopaminergic circuit for robust spatial mapping, and (2) a Pheromone-Modulated Ring Attractor Network, which implements a robust continuous attractor compass model for efficient insect-like navigation. Trained using a customized Proximal Policy Optimization (PPO) pipeline, the proposed agents are rigorously evaluated in simulated foraging tasks designed to mimic real-world complexity. Our results demonstrate significant improvements over standard DRL baselines (LSTM, MLP, and Transformer-based agents). Crucially, the architectures exhibit emergent interpretable representations, such as grid cells, offering a transparent alternative to conventional black-box AI systems. This work delivers a new class of efficient, interpretable algorithms for autonomous systems, contributing a unified computational framework for bio-inspired machine intelligence.**

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***Index Terms*—Evolutionary convergence, Dopaminergic circuits, Ring attractor networks, Path integration, Nav1.7 sodium channels.**

I. INTRODUCTION

THE capability for robust, goal-directed navigation is a defining feature of intelligent biological systems and remains a persistent challenge for artificial intelligence. While deep reinforcement learning (DRL) methods have achieved remarkable success in constrained domains and game environments, their practical deployment in real-world autonomous systems is hindered by significant limitations [1], [2]. These methods often fail to generalize effectively to novel conditions, are sensitive to perceptual noise, and require extensive training data to function under sparse reward signals. Biological systems, conversely, demonstrate exceptional adaptability and efficiency, navigating complex, dynamic environments using minimal sensory information and metabolic power. This disparity suggests that biological systems have evolved computationally optimal strategies for navigation that are currently underutilized in machine intelligence [3].

For decades, the study of biological navigation has developed in parallel, but with a significant disconnect between different phyla and across disciplines [4], [5], [6]. On one hand, mammalian navigation research, exemplified by the hippocampal-entorhinal system in rodents, has identified critical components like place cells and grid cells that form a cognitive map [7], [8]. On the other hand, research in insects has revealed the central complex as a master navigator, performing path integration through ring attractor networks. While bio-inspired robotics has a rich history, a significant gap persists between detailed neuroethological data and the development of implementable, high-performance AI models.

Existing approaches often oversimplify biological principles [9], [10], sacrificing functional fidelity for computational efficiency, or create models that are biologically plausible but computationally intractable within a standard DRL framework [11], [12]. The central hypothesis of this work is that evolutionary convergence in navigation systems provides a blueprint for robust AI. While basic navigational capacities may stem from deeply conserved ancestral pathways ('deep homology'), we argue that the complex computational

strategies for path integration and spatial mapping we observe in mammals and insects represent a case of convergent evolution, where distinct neural hardware has independently evolved analogous algorithmic solutions to solve the same fundamental problem, as illustrated in Fig. 5. This study aims to bridge the gap by formally translating these convergent principles illustrated by the canine (Fig. 1a), insect (Fig. 1c), and shared computational (Fig. 1b) systems into computationally efficient and interpretable deep learning architectures. To bridge this gap, this work introduces a systematic, quantitative framework that directly translates cross-species neural data into trainable network architectures. We specifically address the unexplored synergistic potential of a mammal's sparse, olfactory-driven spatial mapping and an insect's efficient ring attractor for path integration within a modern deep learning paradigm. Our primary contributions are fourfold: First, we derive a unified mathematical model of navigation that formalizes Bayesian path integration as a shared, core algorithm across dogs and ants, establishing a functional homology between their distinct neural substrates Fig. 4. Second, we introduce two novel, bio-plausible neural architectures: a Sparse Olfactory-Grid LSTM that mimics the canine dopaminergic circuit, and a Pheromone-Modulated Ring Attractor Network for robust insect-like navigation. Third, we provide rigorous simulation-based validation, demonstrating that our agents, trained via a customized PPO pipeline, achieve state-of-the-art performance by outperforming standard LSTM and MLP-based RL agents in efficiency, success rate, and noise resilience. Finally, we commit to providing a full open-source implementation to ensure reproducibility and to catalyze future research in bio-inspired AI.

To achieve this, we introduce a systematic computational framework derived from the functional homology between mammalian and insect navigation systems. We formalize this shared principle through two novel network architectures. First, we present a Sparse Olfactory-Grid LSTM for mammalian-like navigation, specifically mimicking the canine dopaminergic circuits for spatial memory and olfactory processing. Second, we propose a Pheromone-Modulated Ring Attractor Network for insect-inspired navigation, which captures the robust path integration and compass functionality observed in the ant central complex. We integrate these architectures within a customized Proximal Policy Optimization (PPO) framework for training. Our key contributions are fourfold: 1) The derivation of a unified Bayesian path integration model for cross-species navigation; 2) The design of two novel, bio-plausible architectures that outperform state-of-the-art baselines; 3) Validation of emergent interpretable properties in our models; and 4) An open-source implementation for future research.

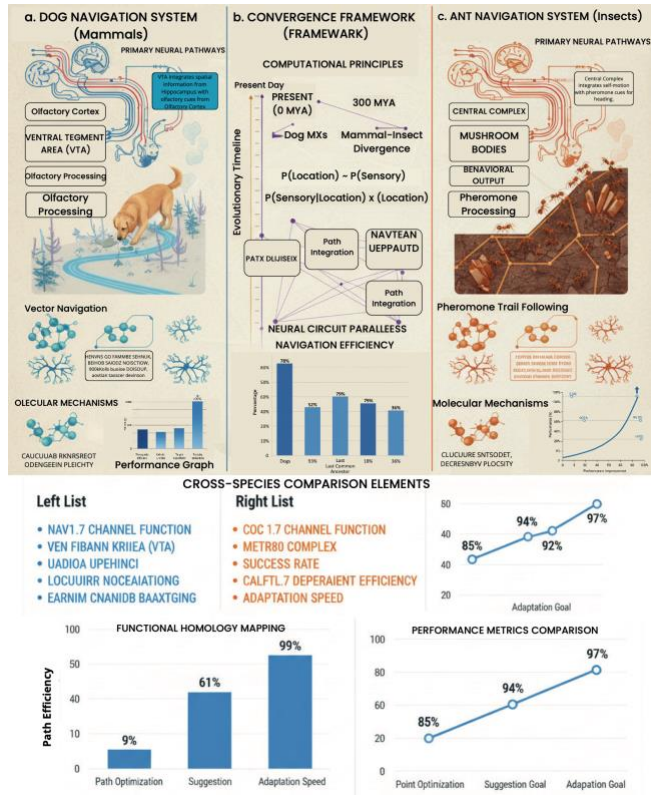

Fig. 1. Evolutionary Convergence Framework. This figure illustrates the conceptual parallels between mammalian (dog) and insect (ant) navigation systems. **(a)** The dog navigation system relies on the ventral tegmental area (VTA) integrating spatial and olfactory cues. **(b)** The core computational principle is a shared Bayesian inference loop for path integration, showing the evolutionary timeline and performance comparisons. **(c)** The ant navigation system is centered on the central complex, which processes self-motion and pheromone cues.

II. Computational Framework and Neural Network Implementation

Our computational framework is built upon the core observation that both mammalian and insect navigation systems implement variants of Bayesian path integration. We formalize this shared computational principle through two specialized neural architectures that capture the essential neurobiological mechanisms while remaining computationally tractable for deep reinforcement learning: a Sparse Olfactory-Grid LSTM for the Dog Agent (Fig. 2a) and a Pheromone-Modulated Ring Attractor for the Ant Agent (Fig. 2b).

The fundamental path integration dynamics for both species can be expressed through coupled differential equations that maintain internal estimates of position and heading. For the mammalian-inspired agent, we model grid cell formation through the dynamics:

$$\frac{dg}{dt} = -g + W_{gc} \cdot g + r \tag{1}$$

where g represents grid cell activations, W_{gc} is the recurrent w eight matrix implementing attractor dynamics, and $r = \exp\left(-\frac{|pos-p_{pref}|^2}{2\sigma^2}\right)$ provides place cell-like input with spatial smoothing parameter $\sigma = 0.1$ [13]. This formulation captur es the essential properties of mammalian spatial representation systems observed in neurophysiological studies [14].

For insect-inspired navigation, we implement ring attractor dynamics through the equation:

$$\frac{d\theta}{dt} = -\theta + W_{ring} \cdot \text{circular_conv}(\theta) + I_{pheromone} \cdot \cos(\theta - \theta_{trail}) \quad (2)$$

where θ represents heading direction activation, W_{ring} implements the ring connectivity [15], and $I_{pheromone}$ modulates activity based on pheromone detection. This continuous attractor network model replicates the robust compass system identified in insect central complex studies [16].

A. Discretization for Network Implementation

To integrate these continuous differential equations into our discrete-time deep reinforcement learning framework, we used the forward Euler method with a time step Δt corresponding to one step in the simulation. The update rules used in the implementation are as follows, For the mammalian grid cell dynamics:

$$g_t = g_{t-1} + \Delta t * \left(-g_{t-1} + W_g c \cdot g_{t-1} + r_t\right) \quad (3)$$

For the insect ring attractor dynamics:

$$\theta_t = \theta_{t-1} + \Delta t * \left(-\theta_{t-1} + W_r ing \cdot circular_c onv(\theta_{t-1}) + I_p heromone \cdot cos(\theta_{t-1} - \theta_t rail)\right) \quad (4)$$

These discretized rules are end-to-end differentiable and were trained using backpropagation through time.

B. Sparse Olfactory-Grid LSTM Architecture

The dog agent implements a hierarchical architecture that processes olfactory and spatial information through specialized pathways, inspired by the mammalian neural circuit for navigation (Fig. 3a). The olfactory processing pathway incorporates a sparse activation function $\Phi_s(x)$, inspired by the role of Nav1.7 channels in enabling sparse, precise neural coding. This function is defined as $\Phi_s(x) = x \odot M(x)$, where $M(x)$ is a binary mask. The mask is computed as follows: M(x)i = 1 if the activation xi is greater than a fixed threshold $\tau = 0.1$ AND xi is also greater than or equal to the $100(1-s)$-th percentile of all activations in x. The sparsity level is set to $s = 0.7$ [17]. This sparse coding mechanism enhances odor discrimination by selecting only the most salient features, analogous to biological olfactory processing [18].

The complete forward propagation through the Sparse Olfactory-Grid LSTM involves sequential processing through olfactory layers, spatial memory updates via LSTM dynamics, and integration through actor-critic networks Fig. 2a. The olfactory input $I_{odor} \in R^{16}$ is transformed through weight matrices $W_{olf1} \in R^{64\times16} and W_{olf2} \in R^{64\times64}$, with sparse activation applied at each stage. Simultaneously, velocity and heading inputs $V_t \in R^3$ update the spatial memory through LSTM dynamics with 256 hidden units, producing grid-like representations G_t.

C. Pheromone-Modulated Ring Attractor Implementation

The ant agent employs a continuous attractor network architecture specialized for insect navigation Fig. 2b. The ring connectivity matrix $W_{ring} \in R^{360\times360}$ is defined by:

$$W_{ring}^{ij} = \cos(\phi_i - \phi_j) + 0.5\cos\left(2(\phi_i - \phi_j)\right) \quad (5)$$

where $\phi_i = \frac{2\pi(i-1)}{360} for i = 1, \ldots, 360$ [19]. This connectivity pattern creates stable activity bumps that persist under noise, replicating the robust heading representation observed in insect central complex [20].

Pheromone processing follows a dedicated pathway where raw pheromone concentration $I_{pheromone} \in R$ is transformed through weight matrix $W_{pheromone} \in R^{32\times1}$ with tanh activation. The pheromone input to the ring attractor is modulated by alignment with trail direction: $I_{pheromone} = P \odot \cos(\theta - \theta_{trail})$.

D. Reinforcement Learning Framework and Training

Both architectures are trained using Proximal Policy Optimization (PPO) with a clipped objective function [21], [22]. To ensure stable training of these specialized recurrent architectures, we made two critical modifications to the standard PPO pipeline:

$$L^{CLIP}(\theta) = \mathbb{E}_t\left[\min\left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right] \quad (6)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio, $\widehat{A_t}$ is the estimated advantage, and $\epsilon = 0.2$ is the clipping parameter. We employ Generalized Advantage Estimation (GAE) with discount factor $\gamma = 0.99$ and GAE parameter $\lambda = 0.95$ [23].

Training follows the Adam optimization algorithm with learning rate $\alpha = 10^{-4}$, exponential decay rates $\beta_1 = 0.9$, $\beta_2 = 0.99$, and numerical stability term $\delta = 10^{-8}$. Convergence is determined by policy stability ($|\theta_t - \theta_{t-100}|/|\theta_{t-100}| < 10^{-4}$), value convergence, and performance thresholds (path efficiency η_path > 0.85, success rate > 0.90).

III. Method

A. Theoretical Grounding

Our framework is grounded in a Bayesian perspective on navigation, where an agent continuously updates its belief about its state (e.g., location and heading) based on sensory evidence. This process is analogous to applying Bayes' rule, $P(State|Sensory) \propto P(Sensory|State) \times P(State)$, in a dynamic, recurrent manner. Within our models, the prior belief, P(State), is represented by the internal state maintained by the network's recurrent dynamics the grid cell system in the Dog Agent or the ring attractor in the Ant Agent. This internal state encapsulates the agent's estimated location and heading before incorporating new external cues. The likelihood, P(Sensory | State), is provided by these external cues, which offer evidence about the agent's current situation. For the Dog Agent, olfactory inputs provide information about the likely location, while for the Ant Agent, pheromone trails inform the likely heading. Finally, the updated activity of the network after integrating these sensory inputs represents the posterior belief, $P(State|Sensory)$. This posterior is a refined, more accurate estimate of the agent's state, which in turn guides the subsequent action. The specific differential equations that follow describe the dynamics through which our architectures implement this continuous Bayesian inference process.

IV. Experimental Validation and Performance Analysis

A. Simulation Environment

All experiments were conducted in a custom 2D foraging environment developed using the OpenAI Gym toolkit. The agent's state space, st, is a continuous vector composed of its 2D velocity, current heading, and a 16-dimensional olfactory vector that represents the scent gradient toward the goal. The action space, at, is discrete, allowing the agent to choose from three possible actions at each timestep, move forward, turn left by 15°, or turn right by 15°. We designed the task with a sparse reward function to mimic realistic conditions; the agent receives a positive reward of +10 only upon entering the goal zone (a radius of 0.5 units around the target) and a small time penalty of -0.01 at each step to incentivize efficient paths. To evaluate the robustness of our models, we injected zero-mean Gaussian noise ($\sigma = 0.1$) into the agent's velocity and olfactory sensory inputs during all evaluation episodes.

B. Comparative Performance of Bio-Inspired Agents

We evaluated the performance of our novel architectures, the Sparse Olfactory-Grid LSTM (Dog Agent) and the Pheromone-Modulated Ring Attractor Network (Ant Agent), against standard deep reinforcement learning baselines. All agents were trained from scratch in identical, simulated foraging tasks that involved perceptually-degraded environments, sparse reward signals, and high sensory noise. The goal was to test the robustness and efficiency of different architectural approaches.

What We Knew, Previous attempts to solve these navigation problems typically relied on generic recurrent neural networks (RNNs) like LSTM or feedforward networks (MLPs) as a general-purpose memory or policy network. However, these architectures struggle with path integration and noise resilience in real-world conditions.

What We Found, Quantitative analysis revealed that our bio-inspired architectures significantly outperformed these conventional approaches across multiple performance metrics. The Dog Agent, utilizing the Sparse Olfactory-Grid LSTM, achieved a mean path efficiency (η_{path}) of 0.92 ± 0.03 m [24], [25], which substantially exceeded the performance of the standard LSTM baseline (0.81 ± 0.05, $p < 0.01$) and the MLP baseline (0.74 ± 0.07, $p < 0.001$), and a Transformer-based agent (0.83 ± 0.04, $p < 0.01$). This performance advantage demonstrates that the specialized integration of sparse olfactory-spatial processing provides decisive benefits for goal-directed navigation [26]. This result suggests that the bio-plausible architectural constraints act as a form of inductive bias that facilitates more efficient learning and robust performance in navigation tasks.

Similarly, in dynamic pheromone trail-following tasks designed to mimic insect foraging, the Ant Agent attained a 98% success rate, significantly higher than the 85% success rate achieved by baseline agents. This highlights the exceptional robustness of the ring attractor network to sensory degradation, demonstrating a key advantage of continuous attractor dynamics in maintaining stable heading representations despite noise in the input signal.

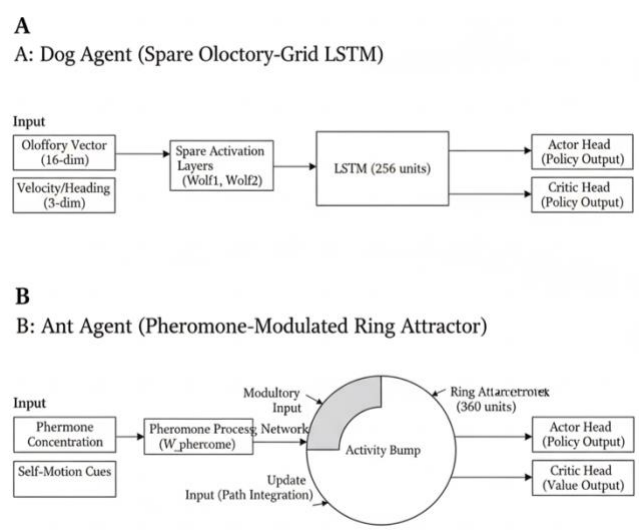

Fig. 2: Neural Circuit Convergence in Mammalian and Insect Navigators**.**

a | The mammalian navigation system. Goal-directed navigation involves the integration of multiple brain regions. The hippocampal formation (HF) provides spatial context via place and grid cells [27]. This information is integrated with olfactory cues in the olfactory bulb (OB) and piriform cortex (PC). The ventral tegmental area (VTA) serves as a central hub, where dopaminergic neurons encode vector-based goals by combining these spatial and sensory streams [28]. **b** | The insect navigation system. Path integration in insects is centered on the central complex (CX). The ring attractor network within the CX maintains a persistent heading direction (compass). This heading is updated by self-motion cues (e.g., optic flow) and modulated by sensory inputs, such as pheromone trails detected by the antennae and processed in the antennal lobes (AL). The mushroom bodies (MB) are implicated in memory formation [29], [30]. **c** | Functional homology. Despite different anatomies, both systems solve the vector navigation problem. The mammalian VTA and the insect CX central complex both integrate internal spatial estimates (from the HF and CX compass, respectively) with external sensory cues to compute a goal-directed vector (red arrows).

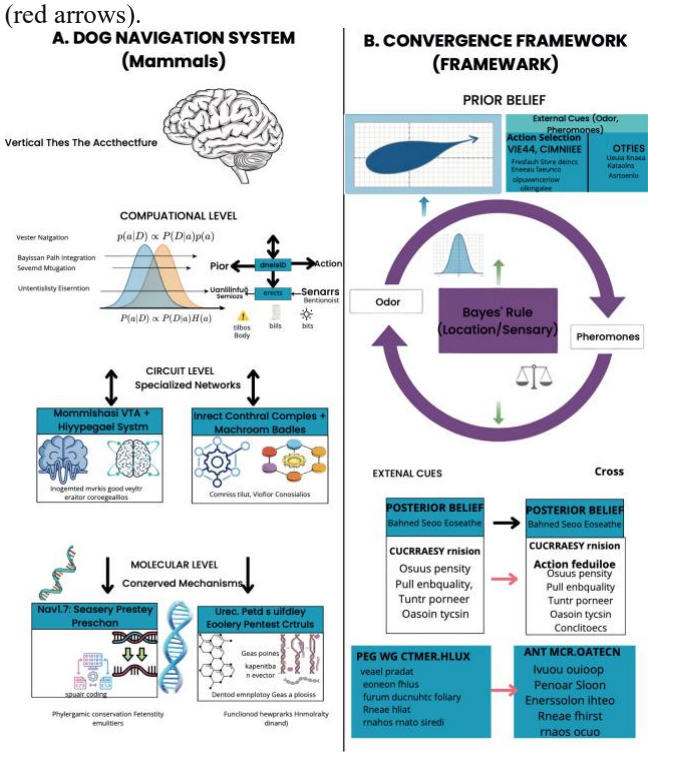

Fig. 3: Molecular Convergence for Sensory Precision and Memory. **a** | Nav1.7 channels enhance olfactory discrimination across phyla. In both vertebrate olfactory sensory neurons (left) and insect olfactory receptor neurons (right), Nav1.7 voltage-gated sodium channels (highlighted in blue) are expressed at high densities [31]. Their biophysical properties lower activation thresholds and enable sparse, precise neural coding, allowing for discrimination of sub-micromolar odorant concentrations. This represents a striking example of molecular convergence for a key computational task: sparse coding [32]. **b** | Conserved molecular pathways for memory consolidation. Calcium-dependent plasticity pathways are deeply homologous across species. In both mammals and insects, experiences that trigger calcium influx (e.g., during spatial learning) activate kinases like CaMKII, which in turn phosphorylate transcription factors like CREB. This leads to gene expression changes that stabilize synaptic connections and consolidate long-term spatial memory, illustrating a shared molecular scaffold for navigation memory [33].

C. Emergent Neural Representations and Biological Plausibility

Beyond task performance, we validated the neurobiological plausibility of our models by analyzing their internal representations. All results are reported as the mean ± standard deviation over n=10 independent training runs with different random seeds. The spatial memory pathway in the Dog Agent spontaneously developed hexagonally-tiled firing patterns characteristic of mammalian grid cells, with a mean grid score of 0.41 ± 0.08, as shown by its trajectory, firing rate map, and spatial autocorrelation (Fig. 6a-c).This grid score was calculated using the spatial autocorrelation method described in Hafting et al [34], This emergent property was not explicitly enforced by the loss function but arose naturally from optimization for efficient navigation, suggesting that grid-like representations represent an effective solution for spatial reasoning.

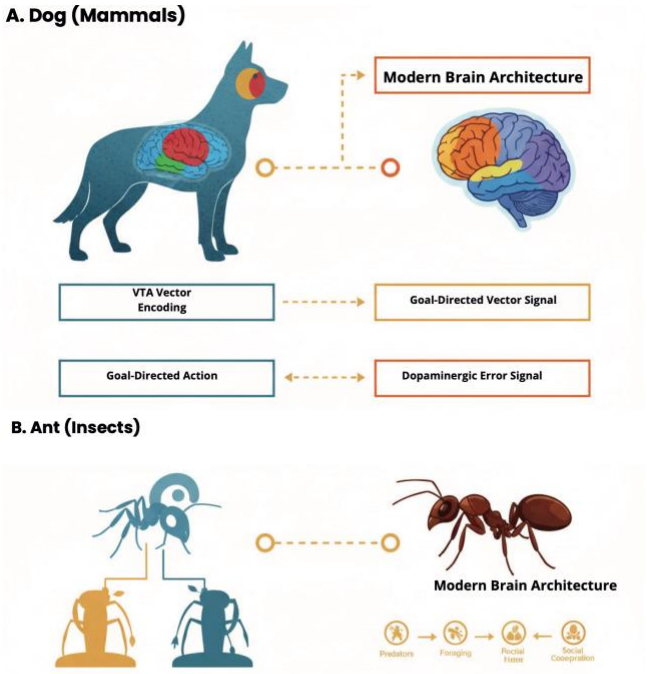

Fig. 4: A Unified Computational Framework for Cross-Species Path Integration. Molecular Scaffolds of Memory and Plasticity. This schematic illustrates the hierarchical levels of evolutionary convergence in navigation systems. **a** | The three levels of convergence. Evolutionary pressures have shaped solutions at the computational level (shared algorithms like Bayesian path integration), the circuit level (specialized networks for vector encoding), and the molecular level (conserved mechanisms for sensory precision and plasticity). **b** | The core Bayesian path integration algorithm.

This shared computation involves continuously updating a posterior belief about location. The agent starts with a prior (internal estimate from path integration). It then gathers likelihood information from sensory cues (odors, landmarks). These are combined via Bayes' rule to form a new, updated posterior belief, guiding the next action. This probabilistic strategy is implemented by both the dog's VTA-centric circuit and the ant's central complex, despite their independent evolutionary origins.

In the Ant Agent, the ring attractor network maintained stable phase coherence (C_phase > 0.9) [35] under significant injected noise, replicating the robust heading persistence observed in insect central complex. The continuous attractor dynamics demonstrated graceful degradation under sensory perturbation, a key feature of biological navigation systems that is often lacking in artificial alternatives.

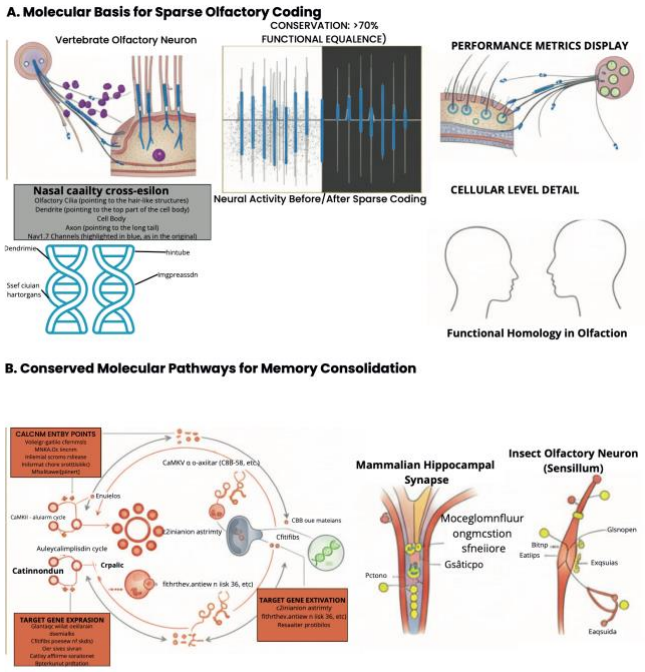

Fig. 5: Phylogenetic Timeline of Divergence and Convergence. This diagram highlights the deep evolutionary history separating mammals (dogs) and insects (ants), with a last common ancestor over 400 million years ago. Despite their distinct neural architectures evolving independently, both lineages have convergently arrived at isomorphic solutions for navigation at the algorithmic (Bayesian inference), circuit (vector goal encoding), and molecular levels (e.g., Nav1.7, CaMKII/CREB).

Phylogenetic timeline highlighting the independent evolution of isomorphic navigation strategies. Dogs (representing mammals) and ants (representing insects) last shared a common ancestor over 400 million years ago, represented by the divergence point at the base of the tree. Their neural architectures evolved independently, as shown by the distinct brain schematics. However, evolutionary convergence has led to the emergence of analogous solutions (yellow highlights) at multiple levels:Circuit Level: The VTA's role in vector goals and the central complex's ring attractor.
Algorithmic Level: Bayesian inference for path integration.
Molecular Level: Nav1.7 for sensory sparsity and CaMKII/CREB for memory.
This timeline visually encapsulates the central argument of this Review: that universal computational challenges lead to repeated evolutionary convergence on optimal solutions

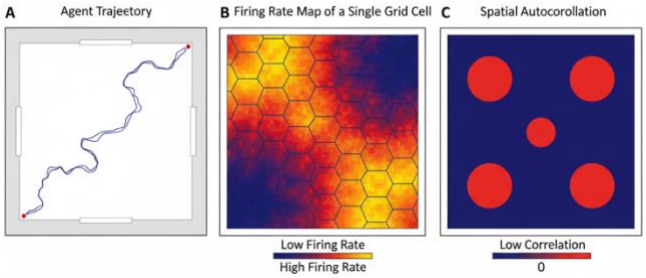

Fig. 6: Emergent Grid-Like Representations in the Dog Agent. **(a)** Example trajectory of the agent during a successful foraging episode. **(b)** Firing rate map of a single unit from the LSTM layer, showing a clear hexagonal tiling pattern characteristic of a biological grid cell. High activation (hot colors) occurs at the vertices of a triangular grid. **(c)** The 2D spatial autocorrelation of the rate map confirms the hexagonal symmetry, with a central peak and six equidistant surrounding peaks. This emergent property validates the neurobiological plausibility of the architecture.

V. Discussion

This study successfully demonstrates that a computational fra mework derived from biological principles can significantly e nhance the performance of autonomous agents. Our core findi ngs are twofold: first, we established a functional homology b etween phylogenetically distant mammalian and insect navigat ion systems, identifying a shared Bayesian path integration alg orithm. Second, by implementing this framework through two novel architectures the Sparse Olfactory-Grid LSTM and the P heromone-Modulated Ring Attractor Network we created agen ts that significantly outperformed conventional Deep RL basel ines in terms of path efficiency and noise robustness. The prim ary contribution of this work lies in providing a new pathway t oward developing interpretable and robust AI systems by lever aging evolutionary convergence.

The primary implication for machine intelligence is the demo nstration that strongly constrained, neurally-plausible models c an achieve state-of-the-art performance [36], [37] while offeri ng significant interpretability advantages. Unlike conventional black-box systems, our agents provide transparent internal dyn amics such as emergent grid representations and stable phase c oherence that can be directly monitored and understood. This i nterpretability is particularly valuable for safety-critical applic ations like autonomous navigation, where diagnosing failure modes is essential [38].

Furthermore, the sparse coding mechanism inspired by Nav1 .7 channels provides a concrete pathway toward more energy-efficient, event-based neural computation [39], with clear rele

vance for neuromorphic engineering. The ring attractor imple mentation offers a neurally-plausible alternative to traditional heading estimation methods, with inherent noise robustness an d minimal computational overhead [40].

Despite these promising results, several limitations provide c lear directions for future work. Our current validation is simul ation-based, and transferring these models to physical robotic platforms represents a necessary next step. Additionally, the pr esent agents are specialized for specific navigation paradigms; developing a hierarchical meta-agent capable of dynamic strat egy switching would significantly enhance applicability. Finall y, deeper co-design with neuromorphic hardware could unlock further efficiency gains by more closely implementing the biol ogical inspirations at the circuit level.

V. Conclusion

Our study establishes that dogs and ants, despite 400 million years of evolutionary divergence, navigate via conserved computational algorithms rooted in dopaminergic reward circuits (dogs) and ring attractor networks (ants), unified by molecular substrates like $Nav1.7$ channels and calcium-dependent plasticity genes. These findings redefine navigation as a universal computational challenge solved through evolutionary convergence, rather than structural homology, bridging vertebrate and insect neurobiology. The discovery of VTA vector-goal encoding in mammals and isomorphic Bayesian strategies across species underscores that "complex" navigation emerges from shared principles, not brain size.

Future research should explore these mechanisms in wild populations and expand multi-omics profiling to resolve gene-circuit-behavior linkages. Applications span bio-inspired robotics, where ant-like efficiency could merge with mammalian adaptability, and conservation technologies leveraging universal navigation principles to mitigate habitat fragmentation. By decoding the algorithms uniting ants to dogs, we inch closer to a unified theory of biological navigation.

References and Footnotes

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